



Histopathologic Cancer Classification using Deep Learning (CNN)

Team & Course

Course: Deep Learning in Healthcare

Team Members:

Aditi Shivapurkar UEC2023301

Minal Chaudhari UEC2023307

Srishti Farande UEC2023312

Shruti Gavahane UEC2023314

Project Scope

Binary classification of histopathology image patches using Convolutional Neural Networks — from raw data to evaluated model performance.



Introduction

Histopathology — the microscopic examination of tissue — is the gold standard for cancer diagnosis. This project applies Convolutional Neural Networks (CNNs) to automate binary classification of image patches: Normal or Tumorous.

Task

Binary classification of 96×96 pixel histopathology image patches from the Kaggle dataset.

Goal

Evaluate CNN architectures on real-world medical image data and benchmark performance metrics.

Approach

Design, train, and compare multiple CNN models — from a shallow baseline to a regularized deeper network.

Problem Statement

The Clinical Challenge

Manual histopathological analysis is time-consuming, expensive, and highly subjective — inter-observer variability among pathologists can significantly affect diagnostic outcomes.

- Requires highly trained specialists
- High cognitive load leads to fatigue errors
- Throughput bottleneck in high-volume labs
- Inconsistent results across readers

Our Objective

Build an **automated deep learning pipeline** that classifies histopathology image patches as cancerous or non-cancerous with high accuracy and reliability.

- ❑ The model must generalize well to unseen patches — not just memorize training samples — making regularization and evaluation critical.

Why Deep Learning for This Task?

→ Traditional ML Limitation

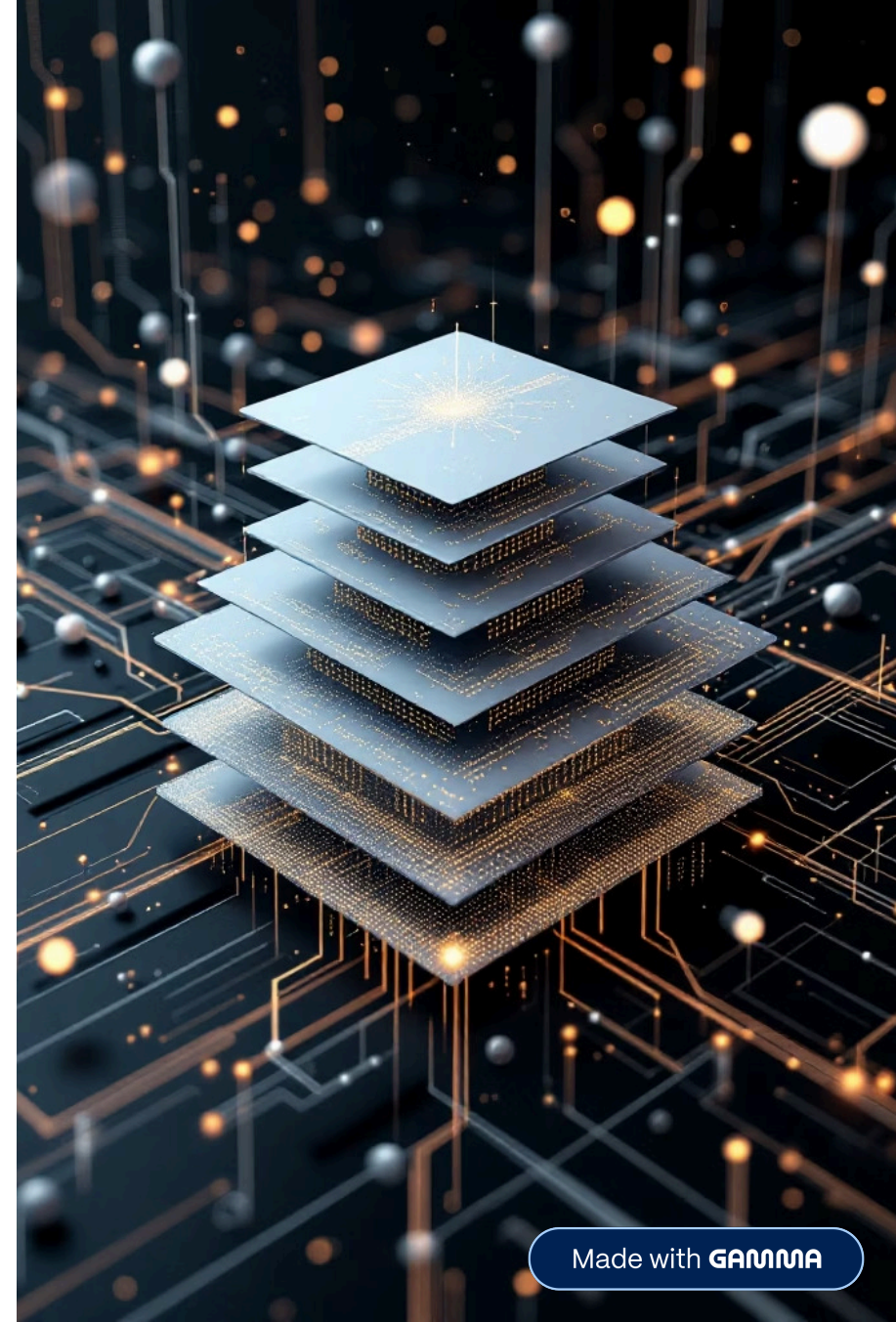
Classical approaches (SVM, Random Forest) require **manual feature engineering** — hand-crafted descriptors like HOG or LBP that may miss subtle pathological patterns.

→ CNN Advantage

CNNs **automatically learn hierarchical spatial features** — edges → textures → tissue structures — directly from raw pixel data through end-to-end training.

→ Medical Imaging Fit

Proven effectiveness in pathology, radiology, and dermatology tasks. CNNs capture local spatial correlations that are diagnostically significant in tissue morphology.



Dataset: Kaggle Histopathologic Cancer Detection

220K

Total Images

Full dataset of 96×96×3 RGB image patches

20K

Sampled

Balanced subset used for efficient training

50/50

Class Split

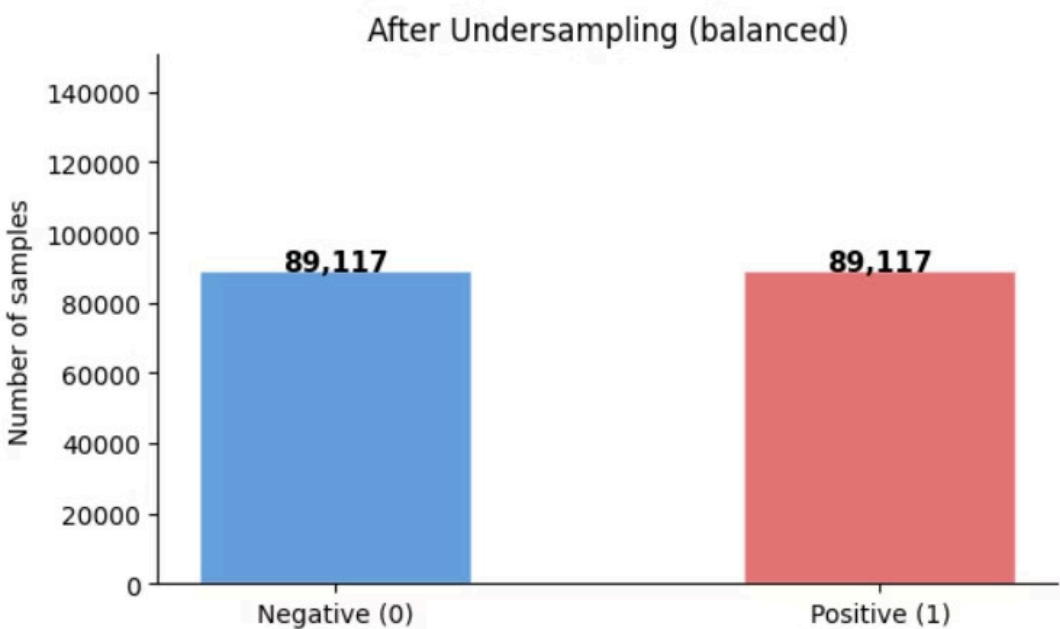
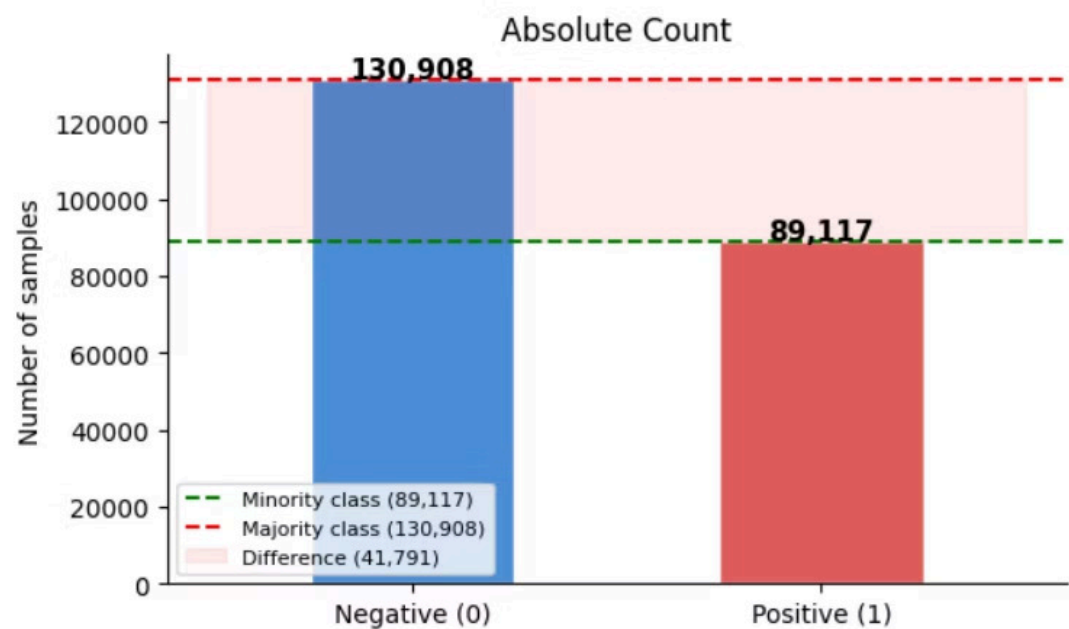
10K positive (cancer) / 10K negative – balanced labels

96px

Patch Size

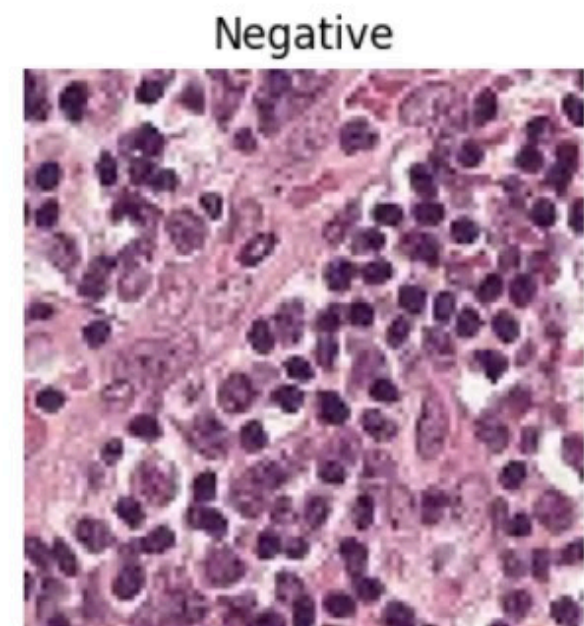
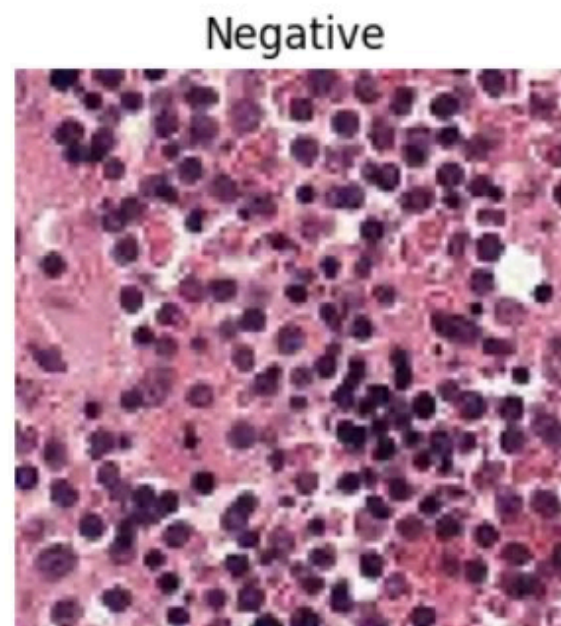
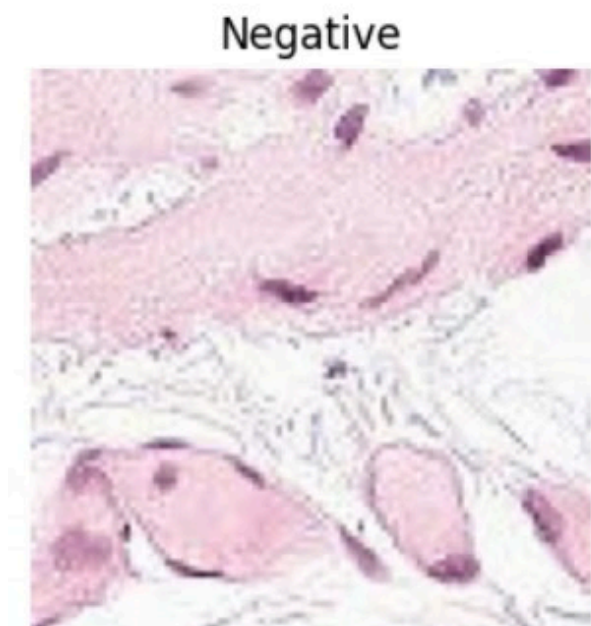
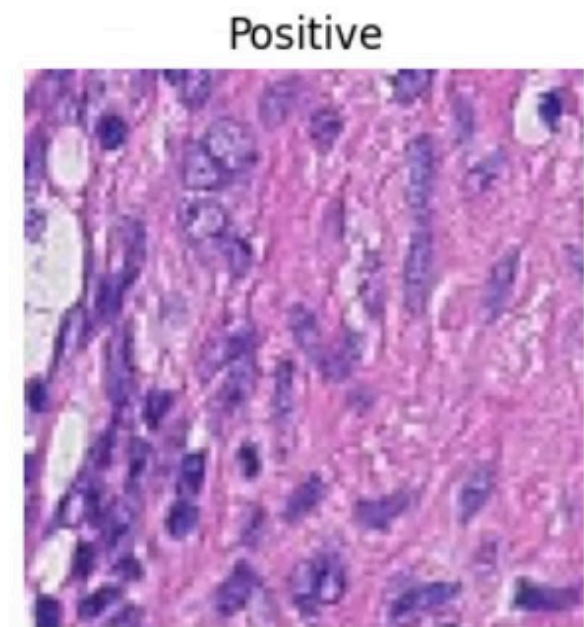
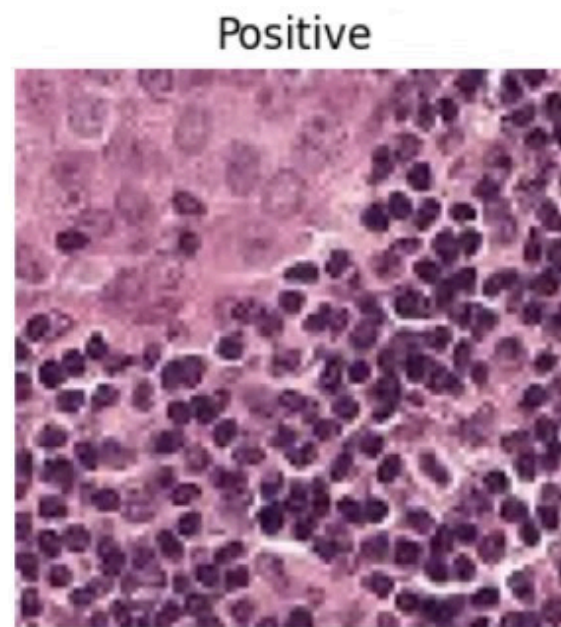
Original resolution; resized to 224×224×3 for model input

PCam — Class Distribution Before Undersampling



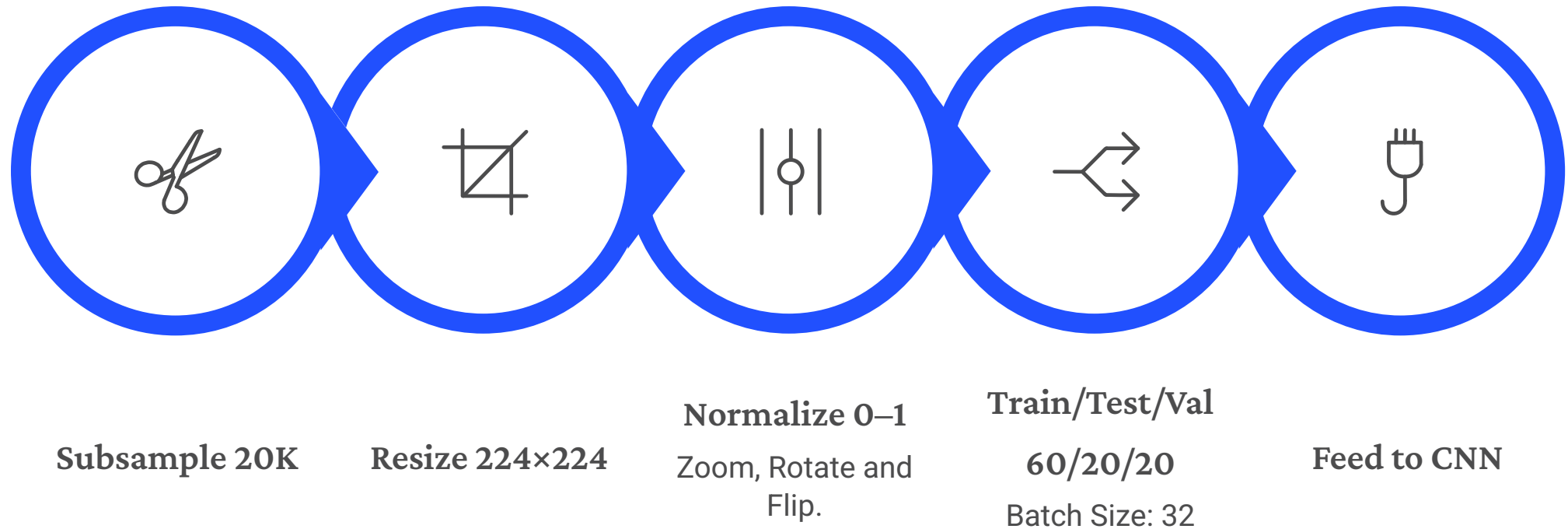
Dataset Sample: Positive & Negative Classes

Visualizing representative image patches from the Kaggle dataset helps understand the subtle differences between cancerous and non-cancerous tissue at a microscopic level.



❏ The visual distinctions can be subtle, underscoring the challenge of manual diagnosis and the need for automated detection.

Data Preprocessing Pipeline



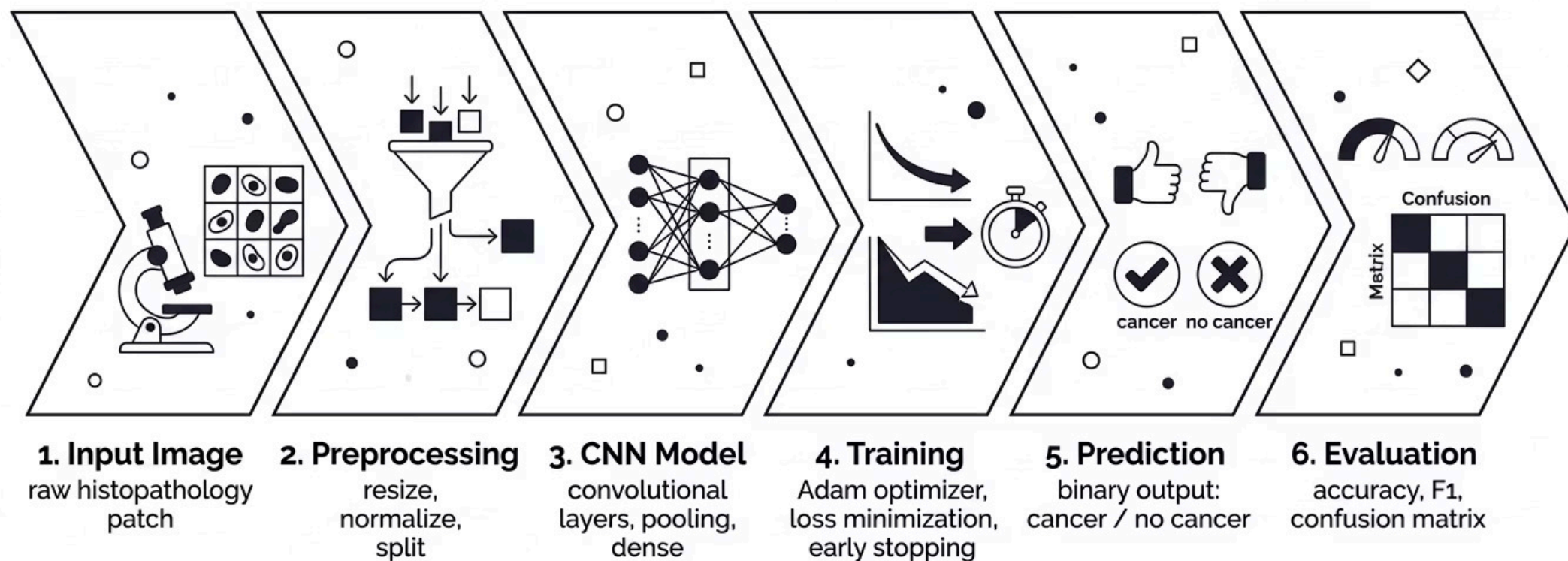
Key Steps

- **Subsampling:** 10K per class for balance
- **Resizing:** 96×96×3 → 224×224×3 (model compatibility)
- **Normalization:** Pixel values scaled to [0, 1]
- **Zoom, Rotate and Flip horizontal and vertical**
- **Split:** 60% train / 20% testing / 20% validation

Why These Choices?

224×224 is the standard input size for many CNN architectures. Normalization ensures stable gradient flow during training. Stratified splitting preserves class distribution in both subsets.

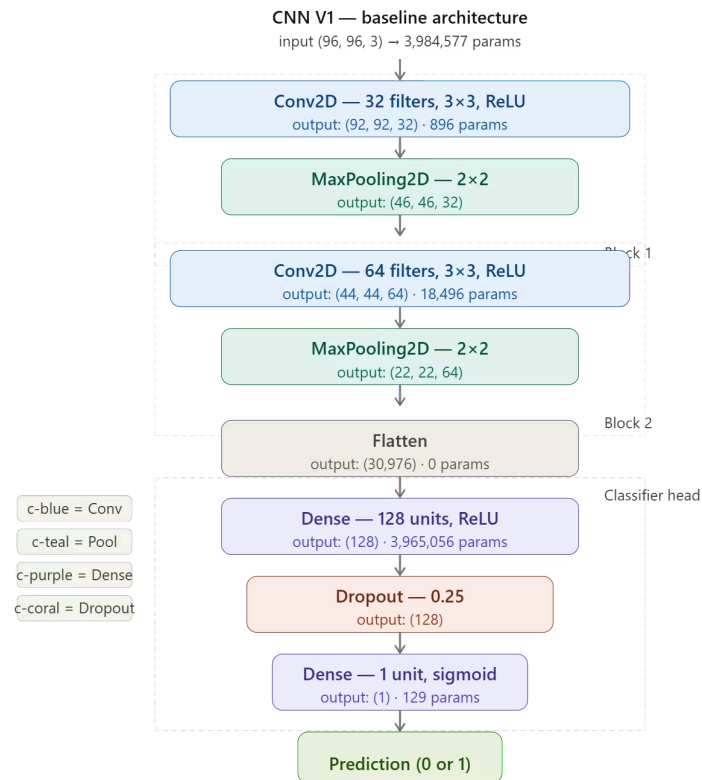
System Pipeline Overview



The pipeline flows from raw image ingestion through preprocessing and model training to final binary prediction and performance evaluation. Each stage is modular – preprocessing and model components can be swapped independently for experimentation.

Model Design — CNN V1 (Baseline)

Purpose & Characteristics

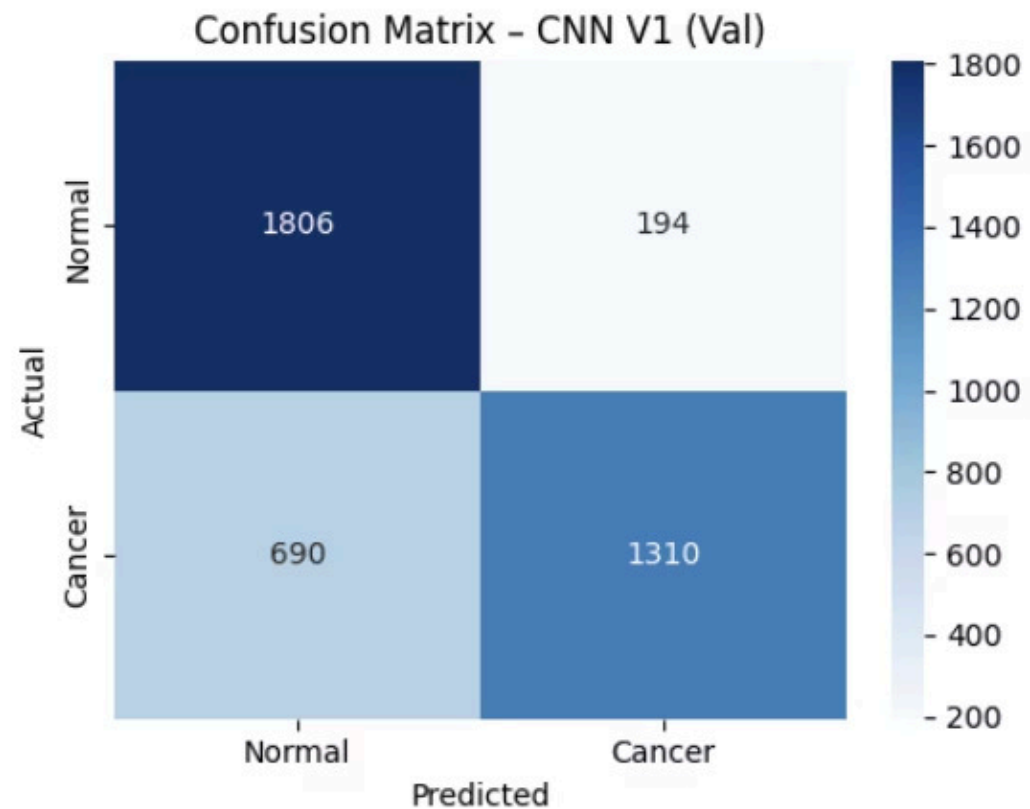


CNN V1 serves as the **reference baseline**. It is intentionally shallow to establish a performance floor before adding complexity.

- Minimal regularization — single Dropout(0.25) at head, no BatchNorm
- ~39–43 seconds per epoch (106ms/step, 375 steps)
- Unstable validation accuracy due to no BatchNorm — oscillates across epochs
- Stopped early at epoch 7 of 10 via EarlyStopping (patience=3)
- Binary cross-entropy loss

📌 Baseline accuracy: ~77.9% — a starting point for a simple architecture.

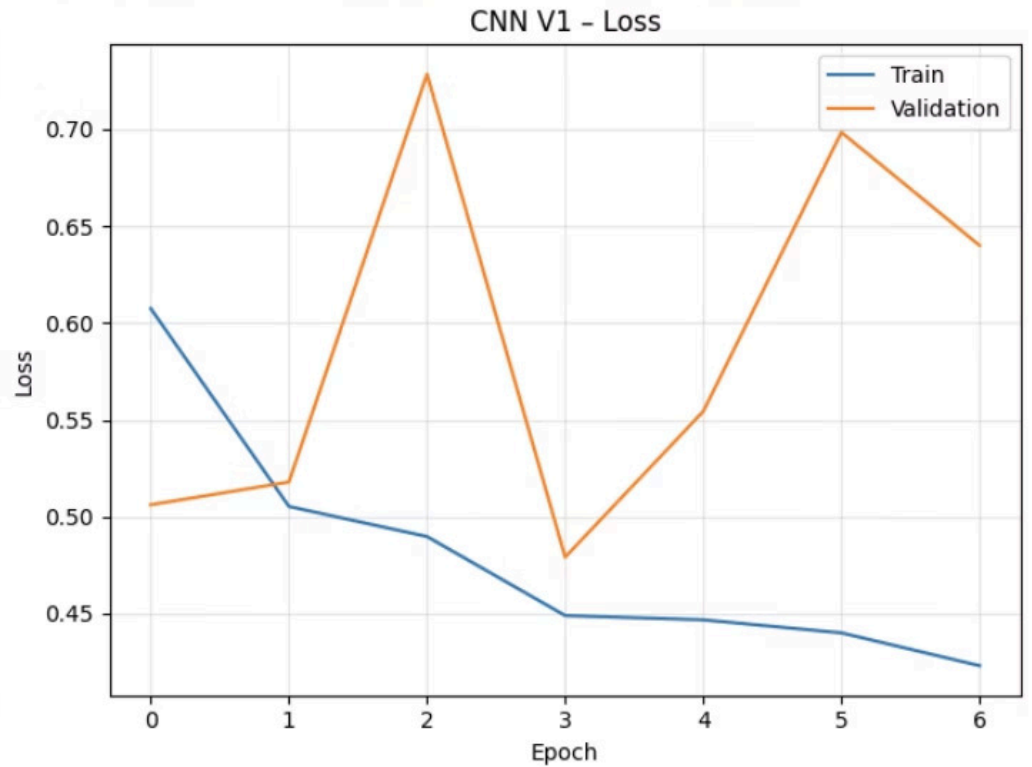
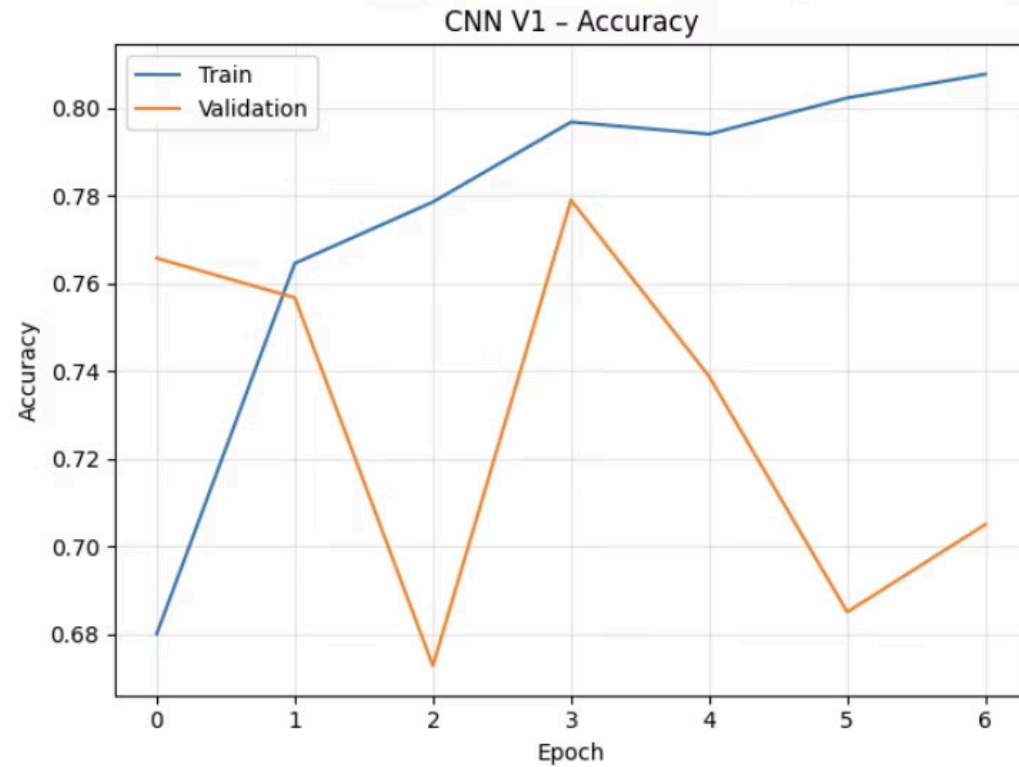
CNN V1 Analysis



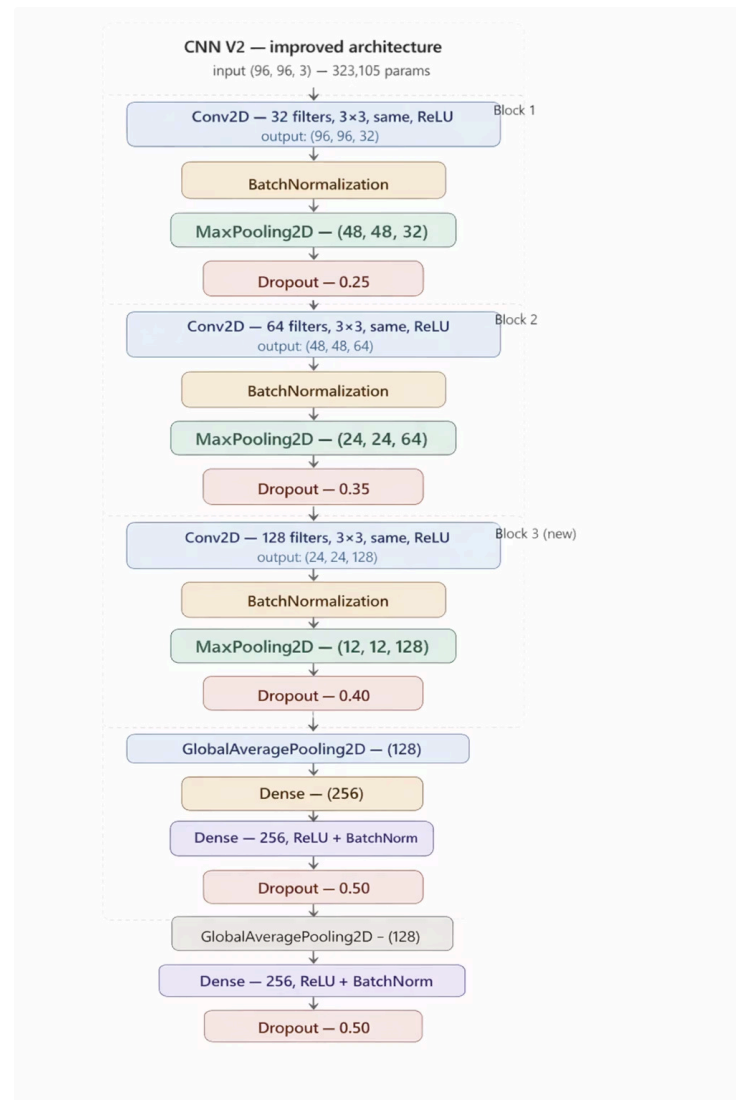
Classification Report:

	precision	recall	f1-score	support
Normal (0)	0.72	0.90	0.80	2000
Cancer (1)	0.87	0.66	0.75	2000
accuracy			0.78	4000
macro avg	0.80	0.78	0.78	4000
weighted avg	0.80	0.78	0.78	4000

CNN V1 Training Curves



Model Design — CNN V2 (Improved)



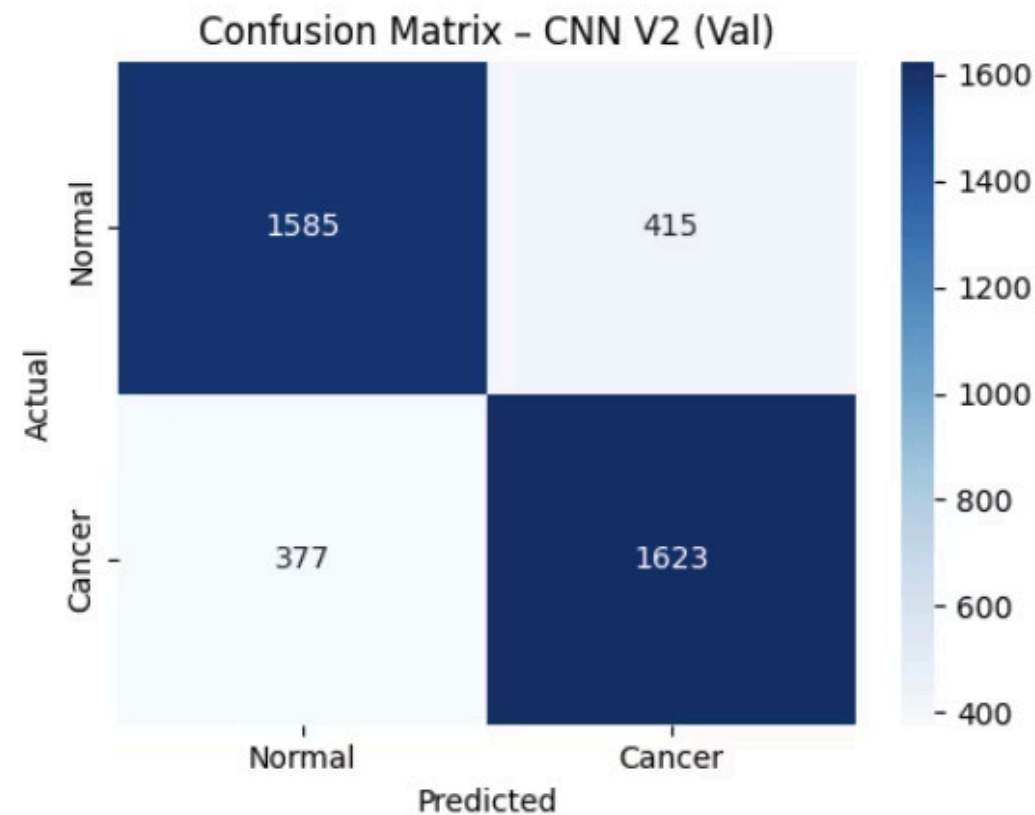
Architectural Improvements

- **Additional Conv2D layers (128 filters)** — the purple-bordered "new block 3" on the right has two extra Conv2D layers with 128 filters that simply don't exist in V1
- **BatchNorm after each Conv** — every Conv in V2 is immediately followed by an amber BatchNormalization box; V1 has none
- **Dropout 0.4–0.5 before Dense** — V1 has only a single 0.25 dropout at the very end; V2 escalates through 0.25 → 0.35 → 0.40 in conv blocks and 0.50 before the final Dense
- **GlobalAveragePooling instead of Flatten** — V1 flattens to 30,976 features; V2 pools to just 128, cutting parameters from ~4M to ~323K
- **Dense(256) → Dense(1)** — V1 uses Dense(128); V2 uses Dense(256) for a richer representation before the sigmoid output

☐ CNN V2 achieved the test accuracy of ~81.1% — demonstrating that moderate depth with regularization outperforms both extremes

Model	Val Acc	Test Acc	ROC-AUC	Recall
CNN V1	0.7050	0.7790	0.8781	0.78
CNN V2	0.7918	0.8107	0.8942	0.81

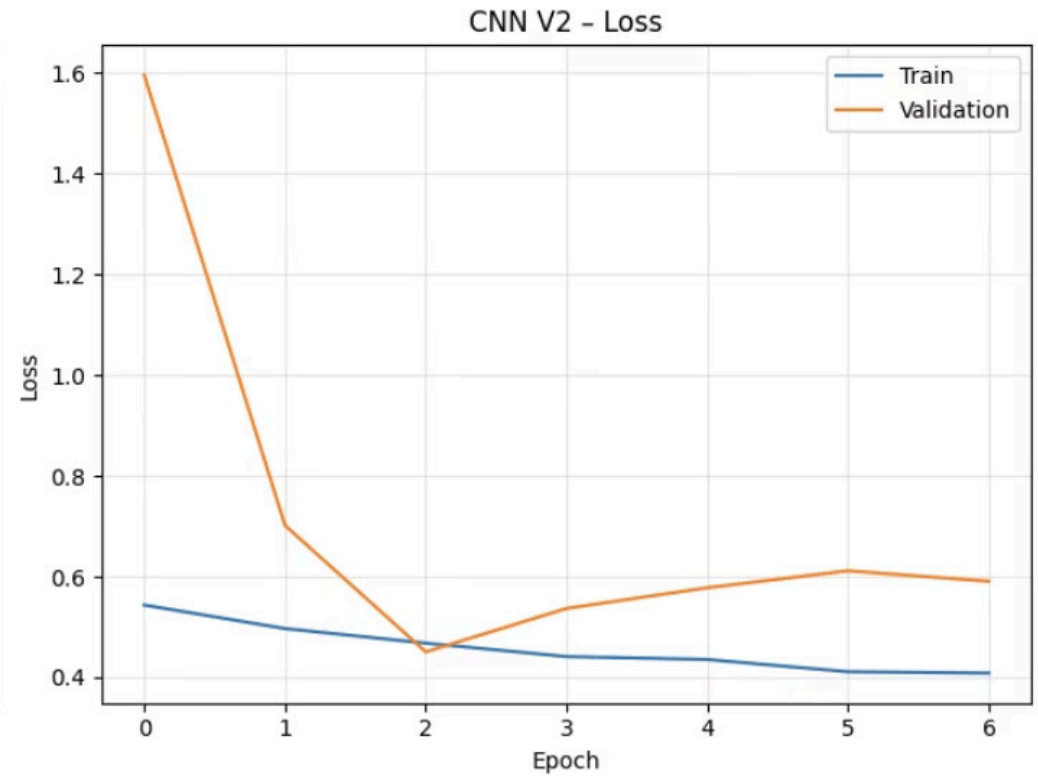
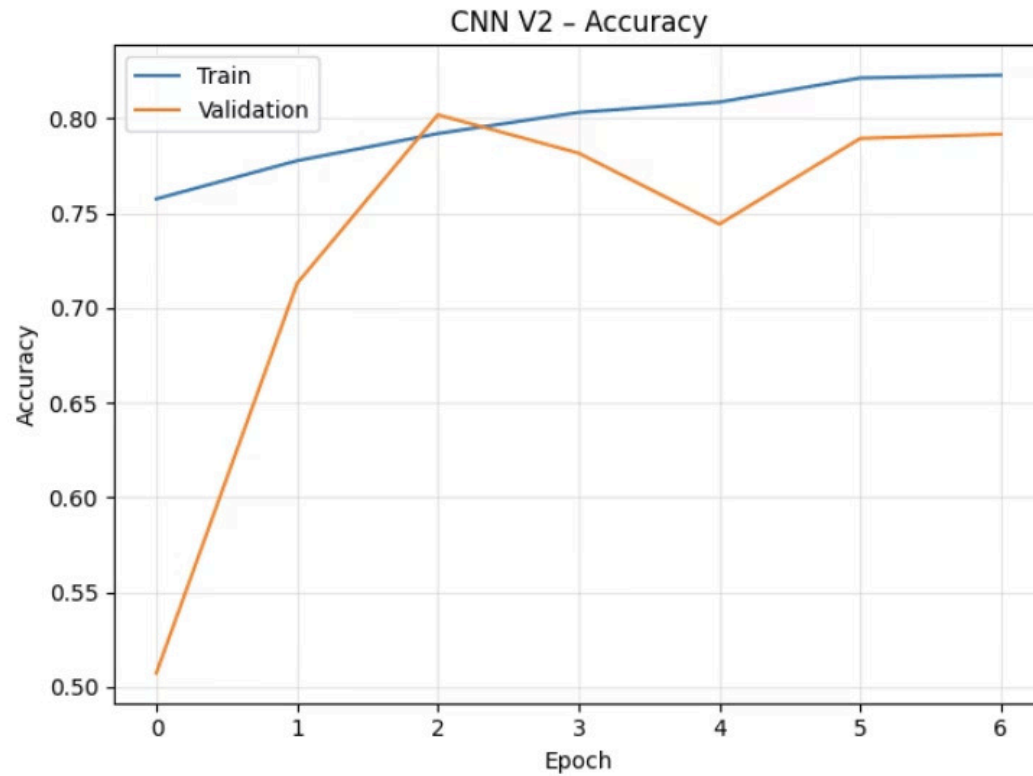
CNN V2 Analysis



Classification Report :

	precision	recall	f1-score	support
Normal (0)	0.81	0.79	0.80	2000
Cancer (1)	0.80	0.81	0.80	2000
accuracy			0.80	4000
macro avg	0.80	0.80	0.80	4000
weighted avg	0.80	0.80	0.80	4000

CNN V2 Training Curves



Hyperparameters & Optimization Strategy

1

Optimizer

Adam — adaptive learning rate optimizer. Combines momentum and RMSProp benefits. Well-suited for sparse gradients in image tasks.

2

Learning Rate

1e-3 for V1 (faster convergence); **1e-4** for V2 (finer tuning, avoids overshooting minima).

ReduceLROnPlateau — reduces LR by factor 0.5 when val_loss plateaus for 2 epochs. Prevents stalling at a suboptimal minimum.

3

Batch Size & Epochs

Batch size: 32 — balance between memory efficiency and gradient stability.

V1: max 10 epochs | **V2:** max 15 epochs — with early stopping to prevent overfitting.

4

Early Stopping

Monitors **val_accuracy** (not val_loss). Patience = **3** for V1, **4** for V2. Restores best weights automatically — prevents overfitting without manual epoch tuning.

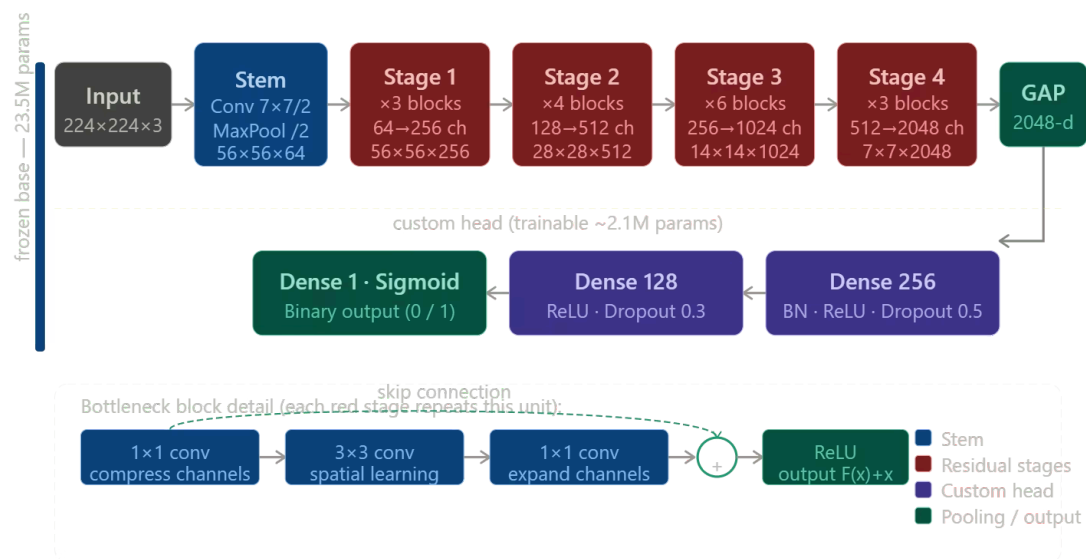
ResNet50 Architecture

ResNet-50 is a 50-layer deep convolutional neural network that introduced skip connections to solve the vanishing gradient problem, enabling stable training of very deep networks. Pretrained on ImageNet's 1.2 million images

How it works:

Residual blocks: Each block uses three convolutions ($1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$). The first compresses channels, the 3×3 learns spatial patterns, the last expands back — efficient and powerful.

Skip connections: A shortcut path adds the input directly to the output ($F(x) + x$). This lets gradients flow freely, preventing them from vanishing in deep layers.



Hyperparameters & Optimization Strategy

1

Optimizer

Adam — adaptive learning rate optimizer. Well-suited for sparse gradients in image tasks.

Activation Function (ReLU)- ReLU kills negative values, keeps positives unchanged. It's computationally cheap, doesn't suffer from the vanishing gradient problem.

2

Learning Rate

1e-3 for Feature Extraction (faster convergence); **1e-5** for Fine Tuning (finer tuning, avoids overshooting minima).

ReduceLROnPlateau — reduces LR by factor 0.5 when val_loss plateaus for 3 epochs. Prevents stalling at a suboptimal minimum.

3

Batch Size & Epochs

Batch size: 32 — balance between memory efficiency and gradient stability.

Feature Extraction: max 10 epochs | **Fine Tuning:** max 15 epochs — with early stopping to prevent overfitting.

4

Early Stopping

Monitors **val_auc**. Patience = **5** for both . Restores best weights automatically — prevents overfitting without manual epoch tuning.

Loss(Binary cross Entropy)-Natural loss for sigmoid outputs because it directly measures the difference between predicted probability and the true 0/1 label.

ResNet50: Training Strategy & Performance

Two-Phase Training Strategy

1

Phase 1: Feature Extraction

ResNet backbone frozen. Only the custom classification head is trained for 10 epochs with a learning rate 1e-3(0.001). This learns task-specific patterns without disturbing pretrained features.

2

Phase 2: Fine-Tuning

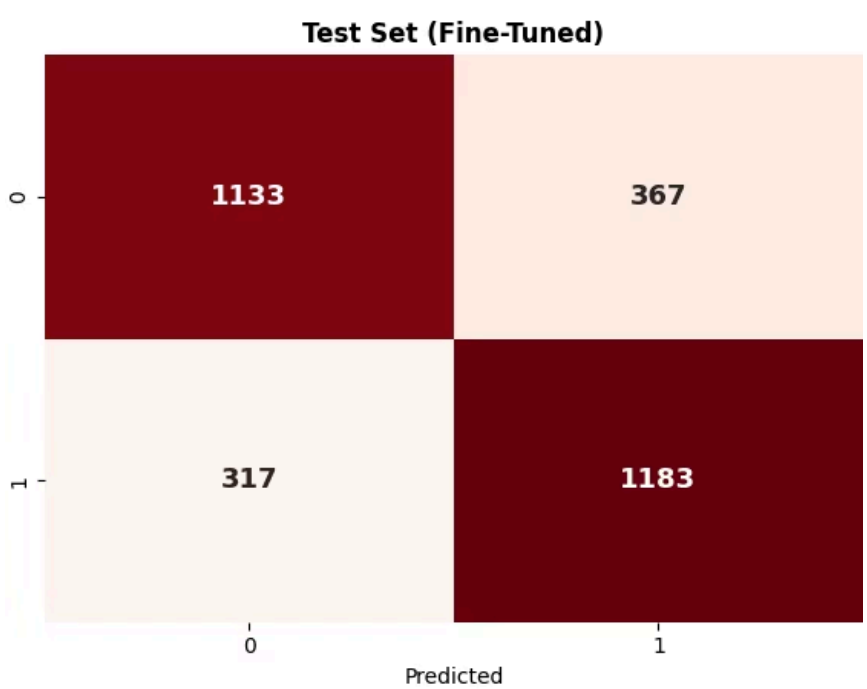
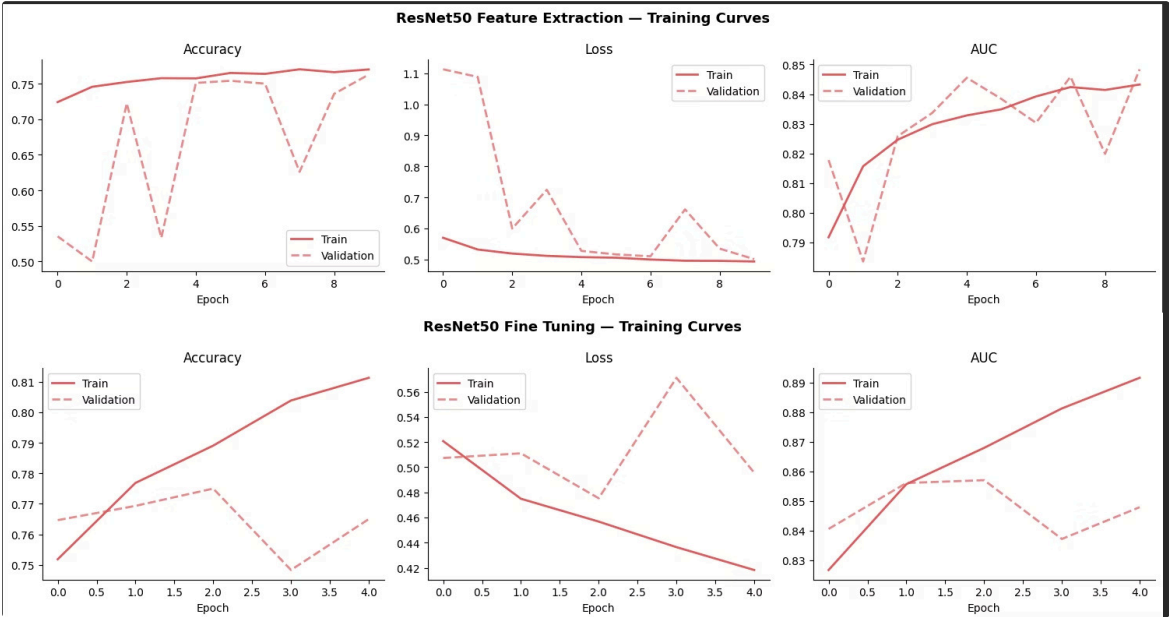
Last 20 ResNetlayers unfrozen. Learning rate reduced to 1e-5(0.00001)for 5 epochs. This allows deeper layers to adapt to specific histopathology features.

Performance Highlights

Metrics	Validation Set	Test Set
Accuracy	0.7613	0.7720
AUC	0.8501	0.8572
Recall	0.6713	0.7887
F1 Score	0.7377	0.7757

Key Insight: This two-phase approach significantly improved performance over pure feature extraction.Fine-tuning achieves higher AUC than feature extraction, indicating better classification capability after adapting deeper layers.

RESNET50 TRAINING CURVES And Confusion Matrix



DenseNet121 Architecture

Why DenseNet-121?

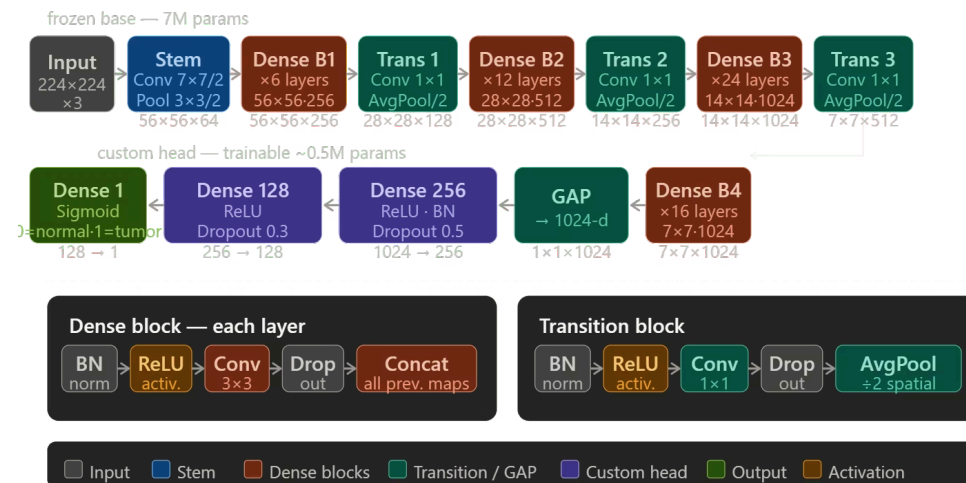
- Dense connectivity: each layer receives feature maps from all previous layers
- Reduces vanishing gradient problem
- Highly parameter-efficient — reuses features instead of relearning them

Architecture Highlights

- Input: 224×224×3 RGB images
- 4 Dense Blocks (×6, ×12, ×24, ×16 layers) with growth rate k=32
- Transition blocks halve spatial resolution and compress channels(÷2)
- Global Average Pooling collapses 7×7×1024 → 1024-d feature vector

Custom Head Design

- FC 1024→256 with ReLU, BatchNorm, Dropout (0.5) — aggressive regularization
- FC 256→128 with ReLU, Dropout (0.3) — fine-grained feature refinement
- FC 128→1 with Sigmoid — probability score for tumor detection



DenseNet121: Training Strategy & Performance

Two-Phase Training Strategy

- 1

Phase 1: Feature Extraction

DenseNet backbone frozen. Only the custom classification head is trained for 10 epochs with a learning rate of 1e-3(0.001). This learns task-specific patterns without disturbing pretrained features.
- 2

Phase 2: Fine-Tuning

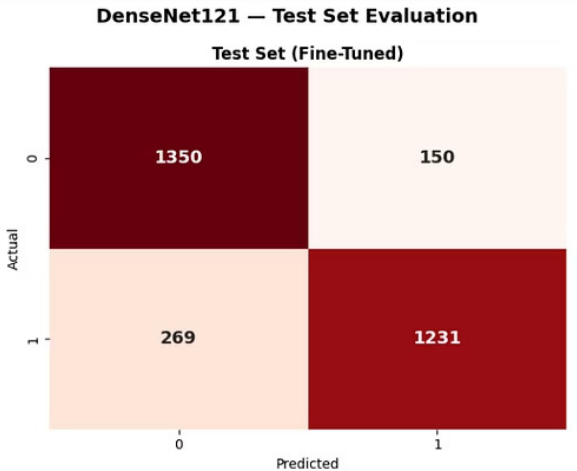
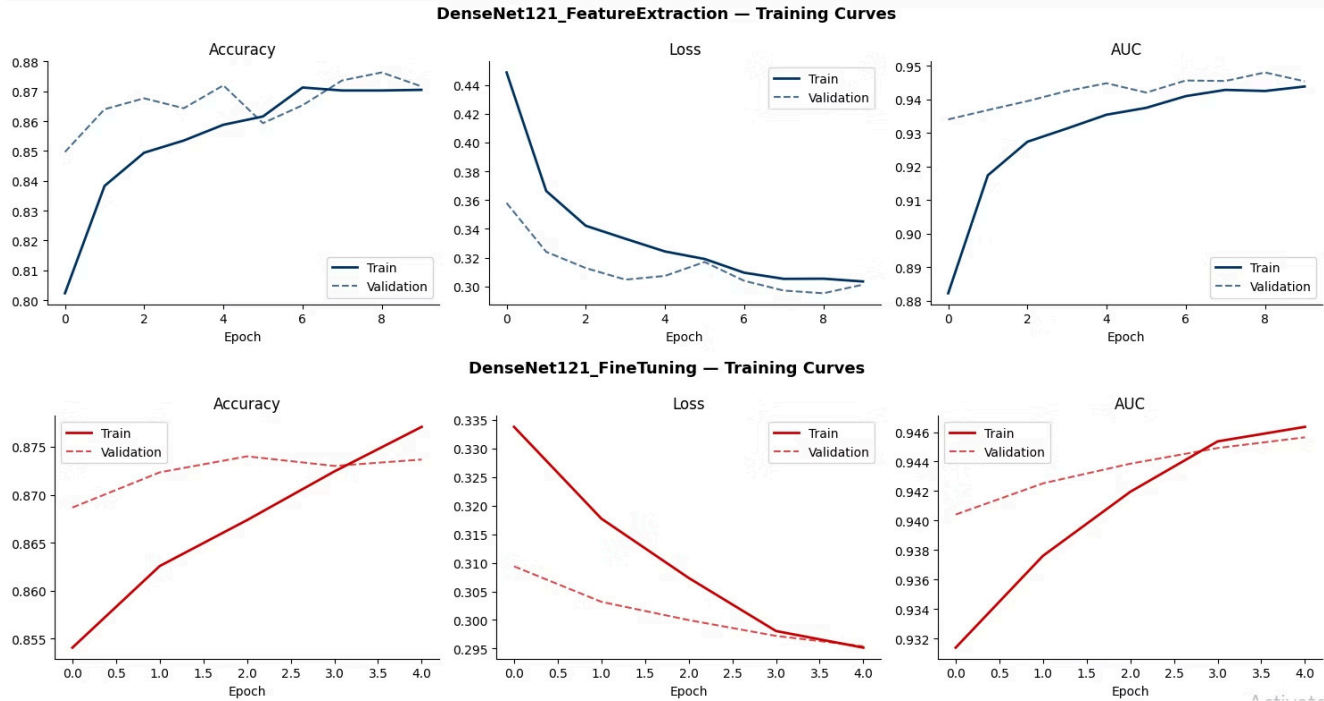
Last 20 DenseNet layers unfrozen. Learning rate reduced to 1e-5 (0.00001)for 5 epochs. This allows deeper layers to adapt to specific histopathology features.

Performance Highlights

Metrics	Feature Extraction	Fine Tuning
Accuracy	0.8767	0.8793
AUC	0.9467	0.9483
Recall	0.8653	0.8633
F1 Score	0.8753	0.8774

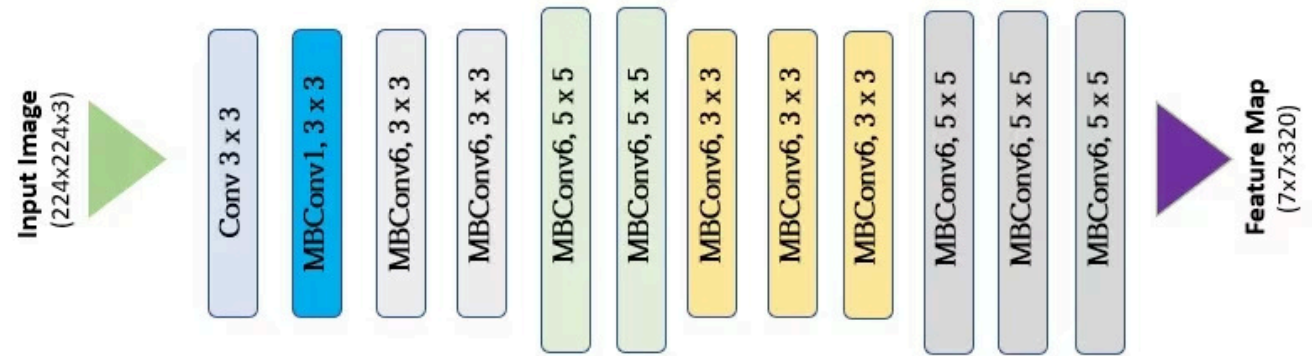
Key Insight: This two-phase approach significantly improved performance over pure feature extraction.Fine-tuning achieves higher AUC than feature extraction, indicating better classification capability after adapting deeper layers.

Training Curves And Confusion Matrix



EfficientNetB0 (Transfer Learning)

- Used **EfficientNet B0** with **pretrained ImageNet weights**
- Model already knows how to detect **basic patterns, edges, textures**
- Removed top layers and added **our own classifier** for cancer / non-cancer
- This saves **training time** and improves **performance**
- Helps a lot because medical images have **complex textures**



Architecture Overview

- EfficientNetB0 Backbone: ImageNet Pretrained
- Input size: $224 \times 224 \times 3$
- Top layers removed

EfficientNetB0: Strategy & Performance

EfficientNetB0 leverages transfer learning and a unique architecture to optimize performance for histopathologic cancer detection:

- **Transfer learning** gives a strong starting point
- **MBConv + SE blocks** help capture **fine tissue textures**
- **Compound scaling** balances depth, width, and resolution
- Trains **faster** due to pretrained weights
- Gives **better generalization** on histopathology images
- Performs better than ResNet for this dataset because it focuses on **important channels and micro-patterns**

EfficientNetB0: Training Strategy & Performance

Two-Phase Training Strategy

1

Phase 1: Feature Extraction

EfficientNet backbone frozen. Only the custom classification head is trained for 10 epochs with a learning rate of 1e-3. This learns task-specific decision boundaries.

2

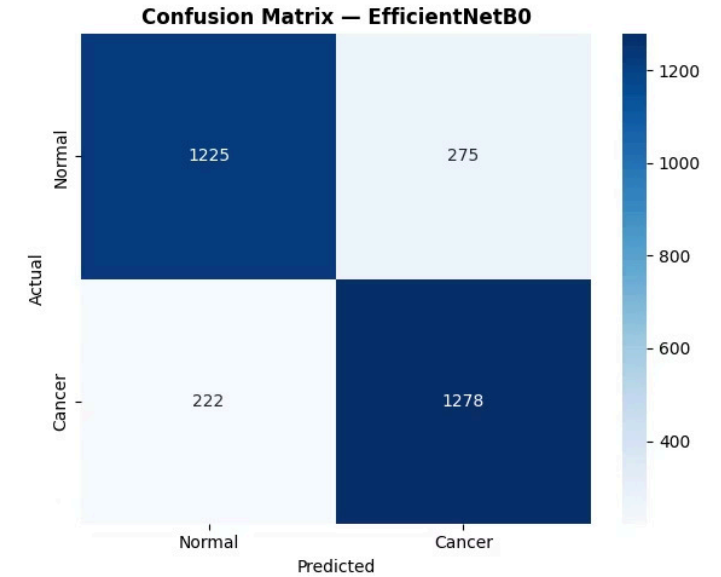
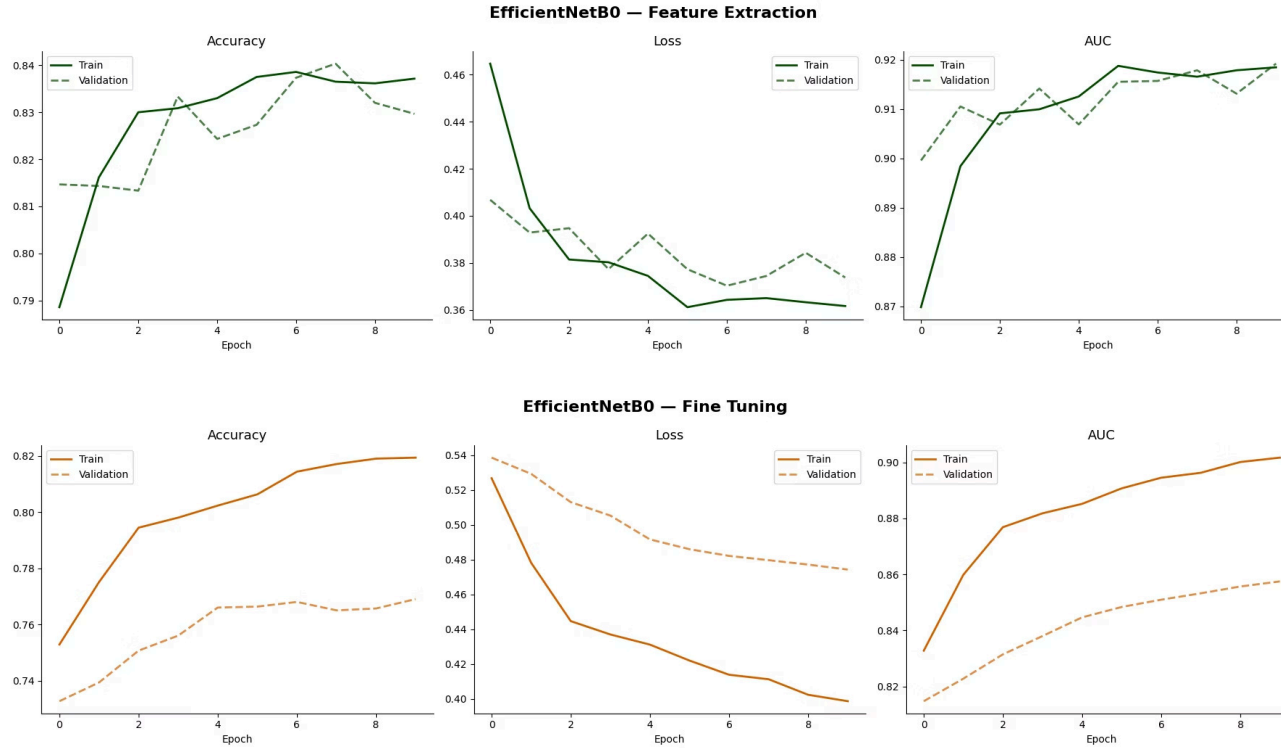
Phase 2: Fine-Tuning

Last 30 EfficientNet layers unfrozen. Learning rate reduced to 1e-5 for 10 epochs. This adapts higher-level convolutional layers to specific histopathology textures.

Performance Highlights

Metrics	Feature Extraction	Fine Tuning
Accuracy	0.8297	0.7690
AUC	0.9192	0.8575
Recall	0.7573	0.7160
Loss	0.3738	0.4743

Training Curves and Confusion Matrix



Evaluation Metrics (Medical AI Focus)

For histopathologic cancer detection, choosing the right evaluation metrics is crucial to ensure clinical utility and patient safety. We prioritize metrics that reflect the model's performance in a medical context.

Accuracy

Overall correctness of predictions (Cancer vs Normal). While intuitive, it can be misleading in imbalanced datasets, where one class significantly outnumbers the other.

AUC (ROC Curve)

Measures the model's ability to distinguish between positive and negative classes across all possible classification thresholds. It's often considered the **most critical metric for medical diagnosis**, as it provides a comprehensive view of classifier performance independent of a specific threshold.

F1-Score

The harmonic mean of Precision and Recall. It provides a balanced assessment, especially important when there's an uneven class distribution. Minimizing false negatives (missing cancer cases) is paramount, making a high F1-score indicative of a reliable model.

❏ **Key Insight:** For histopathology cancer detection, [AUC and F1-score](#) are more important than accuracy due to the critical nature of correctly identifying cancer and handling class imbalances.

Model Performance Comparison

Evaluating the performance of our custom CNN against established transfer learning models on the histopathologic cancer detection task.

Model	Accuracy	AUC	F1 Score
Custom CNN (Baseline)	77%	0.87	0.75
ResNet50 (Pretrained)	77.20%	0.85	0.77
DenseNet121 (Pretrained)	87.93%	0.94	0.877
EfficientNetB0 (Pretrained)	82.70%	0.91	0.81

 **Key Insight:** Our [DenseNet121](#) achieved the highest overall performance. EfficientNetB0 demonstrated competitive results, highlighting the efficacy of transfer learning for specialized medical image analysis.

Result Analysis & Key Insights

Custom CNN

- Learned some dataset-specific textures but had **limited depth and feature reuse**
- Performance was moderate because it could not preserve fine features across many layers

ResNet50

- Focuses more on **global shape learning** through deep residual layers
- Histopathology requires **micro-texture detection**, so performance was limited

DenseNet121

- **Best performance** due to dense connectivity
- Every layer receives features from all previous layers → **excellent texture preservation**
- Captured subtle cancer patterns very effectively (highest AUC & F1)

EfficientNetB0

- Balanced performance with **fewer parameters**
- SE blocks helped focus on **important texture channels**
- Strong generalization and computational efficiency

Key Takeaways

- Histopathology tasks require models that **preserve fine texture information**
- Transfer learning greatly **improves performance** on medical image datasets
- Model architecture choice is **more important than just increasing depth**



Challenges Faced

During the development of the cancer detection model, several significant obstacles were encountered, impacting model performance and development efficiency.

Subtle Data Complexity

Visual distinctions between cancerous and non-cancerous cells are extremely subtle, even for human pathologists, making accurate machine learning difficult.

Limited Image Context

Small 96x96 pixel patches provide insufficient context for identifying crucial diagnostic features like tissue organization and invasion patterns.

Extensive Training Durations

Training deep learning models on large image datasets requires significant computational time, prolonging experimentation and iteration cycles.

Resource Constraints

Limited access to high-performance GPUs slowed down development, restricted model complexity, and hindered extensive hyperparameter tuning.

Medical Image Variability

Inconsistencies in staining, lighting, and tissue appearance across images introduced noise, making it harder for the model to generalize effectively.

Optimization Limitations

Due to time and resource constraints, comprehensive hyperparameter tuning and exploring a wider range of advanced architectures were not fully feasible.

Research Papers

1) Transfer Learning for Histopathology Cancer Detection

<https://arxiv.org/pdf/2112.15523>

2)DenseNet-based Histopathology Classification

[Classification of Breast Cancer Histopathological Images Using DenseNet and Transfer Learning](#)

3)EfficientNet + ResNet Hybrid for Cancer Detection

[Loading...](#)

4) Breast Cancer Image Classification using Custom CNN

https://www.jatit.org/volumes/Vol102No6/21Vol102No6.pdf?utm_source=chatgpt.com